

# Root systems

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## 0.1 Root pairings

### 0.1.1 Definitions

Recall that a *perfect pairing* is a quadruple  $(R, M, N, \mathcal{L})$  where  $R$  is a ring,  $M$  and  $N$  are  $R$ -modules, and  $\mathcal{L} : M \times N \rightarrow R$  is a bilinear map such that the induced maps  $M \rightarrow \text{Hom}_R(N, R)$  and  $N \rightarrow \text{Hom}_R(M, R)$  are isomorphisms of  $R$ -modules.

If  $I$  is a set, denote by  $\text{Perm}(I)$  the group of permutations of  $I$ .

**Definition 1.** A *root pairing* is a tuple  $(R, M, N, \mathcal{L}, I, \alpha, \alpha^\vee, s)$  where  $(R, M, N, \mathcal{L})$  is a perfect pairing,  $I$  is a set,  $\alpha : I \rightarrow M, i \mapsto \alpha_i$ ,  $\alpha^\vee : I \rightarrow N, i \mapsto \alpha_i^\vee$ , and  $s : I \rightarrow \text{Perm}(I), i \mapsto s_i$ , with the conditions that

1.  $\alpha, \alpha^\vee$  are injective,
2. For all  $i \in I$ ,  $\mathcal{L}(\alpha_i, \alpha_i^\vee) = 2$ ,
3. For all  $i, j \in I$ , we have

$$\begin{aligned}\alpha_{s_i(j)} &= \alpha_j - \mathcal{L}(\alpha_j, \alpha_i^\vee) \alpha_i, \\ \alpha_{s_i(j)}^\vee &= \alpha_j^\vee - \mathcal{L}(\alpha_i, \alpha_j^\vee) \alpha_i^\vee.\end{aligned}$$

### 0.1.2 Root pairing of type A

Consider the perfect pairing  $(R, M, N, \mathcal{L}) = (\mathbb{Z}, \mathbb{Z}^n, (\mathbb{Z}^n)^*, (\cdot, \cdot))$  where  $(\cdot, \cdot)$  is the canonical pairing between  $\mathbb{Z}^n$  and its algebraic dual  $(\mathbb{Z}^n)^* := \text{Hom}_{\mathbb{Z}}(\mathbb{Z}^n, \mathbb{Z})$ , i.e. the dot product.

We denote by  $e_1, \dots, e_n$  the standard basis of  $\mathbb{Z}^n$  and by  $e_1^*, \dots, e_n^*$  the dual basis of  $(\mathbb{Z}^n)^*$ .

Write  $[1, n] = \{1, \dots, n\}$ . An *interval* in  $[1, n]$  is a subset of the form  $[i, j] \stackrel{\text{denotes}}{=} \{i, i+1, \dots, j\}$  for some  $1 \leq i \leq j \leq n$ . We denote by  $I_n$  the set of intervals in  $[1, n]$ . Write  $[i] := [i, i]$  for  $1 \leq i \leq n$ .

A *signed interval* in  $[1, n]$  is an element of  $I_n := I_n \times \{1, -1\}$ . We denote by  $I_n^+$  the set of signed intervals of the form  $(J, 1)$  and by  $I_n^-$  the set of signed intervals of the form  $(J, -1)$ . It is clear that  $I_n = I_n^+ \sqcup I_n^-$ . Throughout the rest of this file, we write  $J$  for the signed interval  $(J, 1)$  and  $-J$  for  $(J, -1)$ .

Let  $A = (a_{i,j})$  be the Cartan matrix of type  $A_n$  defined by  $a_{i,i} = 2$  for all  $i \in [1, n]$ ,  $a_{i,j} = -1$  if  $|i - j| = 1$ , and  $a_{i,j} = 0$  otherwise.

Define

$$\begin{aligned}\alpha : I_n &\rightarrow \mathbb{Z}^n, & \forall J \in I_n^+, \quad \alpha_{\pm J} &= \pm \left( \sum_{j \in J} A e_j \right), \\ \alpha^\vee : I_n &\rightarrow (\mathbb{Z}^n)^*, & \forall J \in I_n^+, \quad \alpha_{\pm J}^\vee &= \pm \left( \sum_{j \in J} e_j^* \right).\end{aligned}$$

Finally, define  $s : I_n^+ \rightarrow \text{Perm}(I_n)$  as follows. Fix  $J, K \in I_n^+$ . It is clear that either  $J = K$ ,  $J \subset K$ ,  $K \subset J$ , or and set  $L = (J \cup K) \setminus (J \cap K)$ . It is clear that  $J, K, L$  satisfy exactly one of the following four conditions.

**R1.**  $L = \emptyset$ .

**R2.**  $L \in I_n$  and  $K \subset J$ , and

**R3.**  $L \in I_n$  and  $K \not\subset J$ .

**R4.**  $L \notin I_n \cup \{\emptyset\}$ .

Then,

$$s_J(K) = \begin{cases} -K & \text{in Case \textbf{R1},} \\ -L & \text{in Case \textbf{R2},} \\ L & \text{in Case \textbf{R3}.} \\ K & \text{in Case \textbf{R4},} \end{cases} \quad (1)$$

We extend  $s_J$  to  $I_n$  by setting  $s_J(-K) = -s_J(K)$  for all  $K \in I_n^+$ . We extend  $s$  to  $I_n$  by setting  $s_{-J} = s_J$  for all  $J \in I_n^+$ .

**Lemma 2.** *For all  $J$ , we have  $s_J \in \text{Perm}(I_n)$ .*

*Proof.* Fix  $J \in I_n$ . We prove that  $s_J^2 = \text{id}$  by a straightforward case-by-case check using the definition of  $s_J$  and the four cases above. Then  $s_J^{-1} = s_J$ , and our claim is proved.  $\square$

**Theorem 3.** *The 8-tuple  $(\mathbb{Z}, \mathbb{Z}^l, (\mathbb{Z}^l)^*, \mathcal{L}, I, \alpha, \alpha^\vee, s)$  is a root pairing of  $A_l$ , where:*

1.  $\mathcal{L}$  is the dot product on  $\mathbb{Z}^l \times (\mathbb{Z}^l)^*$
2.  $I = \{\text{signed intervals on the set } \{1, \dots, n\}\} = I^+ \cup I^-$
3.  $s : I \rightarrow \text{Perm}(I)$  where for  $J \in I$ ,  $s_J = s_{-J}$ ,  $s_J(-K) = -s_J(K)$ , defined as  $s_J(K) =$

$$\begin{cases} J \cup K & \text{if } J \cap K = \emptyset \wedge J \cup K \in I \\ K & \text{if } J \cap K = \emptyset \wedge J \cup K \notin I \\ K \setminus J & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge J \subset K \wedge K \setminus J \notin I \\ -K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K = \emptyset \\ -(J \setminus K) & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \in I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \subseteq J \wedge J \setminus K \notin I \\ K & \text{if } J \cap K \neq \emptyset \wedge K \not\subseteq J \wedge J \not\subseteq K \end{cases}$$

*Equivalently (for use in Lean),*

1.  $I = \{(i, j, \varepsilon) | 1 \leq i \leq j \leq l, \varepsilon = \pm 1\}$
2.  $\alpha : I \hookrightarrow \mathbb{Z}^l, (i, j, \varepsilon) \mapsto \varepsilon \left( \sum_{k=i}^j A_l e_k \right)$  (where  $A_l$  is the Cartan Matrix of  $A_l$ )
3.  $\alpha^\vee : I \hookrightarrow (\mathbb{Z}^l)^*, (i, j, \varepsilon) \mapsto \varepsilon \left( \sum_{k=i}^j e_k^\top \right)$
4.  $s : I \rightarrow \text{Perm}(I)$  where  $s(i, j, 1) = s(i, j, -1)$ ,  $s(i, j, \varepsilon)(k, l, 1) = (p, q, \eta)$  means that  $s(i, j, \varepsilon)(k, l, -1) = (p, q, -\eta)$ , and

$$s(i, j, 1)(k, l, 1) = \begin{cases} (k, l, -1) & \text{if } (i, j) = (k, l) \\ (l+1, j, -1) & \text{if } i = k, j > l \\ (j+1, l, 1) & \text{if } i = k, j < l \\ (i, k-1, -1) & \text{if } i < k, j = l \\ (k, i-1, 1) & \text{if } i > k, j = l \\ (k, j, 1) & \text{if } i = l+1 \\ (i, l, 1) & \text{if } k = j+1 \\ (k, l, 1) & \text{otherwise} \end{cases}$$

*Proof.*

□